

Solution

SOUND-05

Class 09 - Science

1. A battery has stored chemical energy. The chemical energy is converted into electrical energy during lighting of bulb. So, a battery converts chemical energy into electrical energy. The filament in the bulb becomes white hot and gives out light. Electrical energy is converted into heat and light energy.

Thus, the energy changes involved in this process can be written as follows:

Chemical energy $\rightarrow$ Electrical energy $\rightarrow$ Heat energy + Light energy.

2. Force = 7 N

Displacement = 8 m

Work done = Force $\times$ Displacement

$$= 7 \times 8 = 56 \text{ J}$$

3. Yes, mechanical energy comprises of both potential energy and kinetic energy. Zero momentum means that velocity is zero.

Hence, there is no kinetic energy but the object may possess potential energy.

4. Given: $m = 4 \text{ kg}$, $F = 16 \text{ N}$, $u = 0$, time $t = 10 \text{ s}$

$$\text{Now, acceleration, } a = \frac{F}{m} = \frac{16}{4} = 4 \text{ ms}^{-2}$$

Therefore, velocity v of the body after 10 s is given by $v = u + at = 0 + 4 \times 10 = 40 \text{ ms}^{-1}$

Hence, K.E. of the body at the end of 10s

$$\text{K.E.} = \frac{1}{2} mv^2 = \frac{1}{2} \times 4 \times (40)^2 = 3200 \text{ J}$$

5. Given, Power of electric iron, $P = 1200 \text{ W} = \frac{1200}{1000} = 1.2 \text{ kW}$

$$\text{Time, } t = \frac{30}{60} = 0.5 \text{ h}$$

Electric energy consumed in a day = Power $\times$ time $= (1.2 \times 0.5) \text{ kWh}$

Electric energy consumed in month of April (i.e. 30 days), $E = (1.2 \times 0.5) \times 30 = 18 \text{ kWh}$.

6. The work done (W) in lifting a box through a distance s against the gravitational force (F) is given by $W = Fs$. Hence it is obvious that it is independent of the rate at which the box is lifted.

7. Mass of substance $m = 50 \text{ g}$

Volume of substance $V = 20 \text{ cm}^3$

Therefore density of substance is

$$D = \frac{M}{V} = \frac{50}{20} = 2.5 \text{ g cm}^{-3}$$

The substance will sink in water, because its density is more than that of water.

8. Suppose a body of mass m is lifted vertically upwards against the gravity (g) through a distance h .

Here, the force (F) required to lift the body = weight of the body, mg (where m is mass and g is acceleration due to gravity).

Now, Work done (W) in lifting a body = Force $\times$ displacement = Weight of body $\times$ Vertical distance

$$W = mg \times h = mgh.$$

Therefore, Work done against the gravity = mgh

9. Given : Force acting on the horse-cart system = 300 N, Initial velocity

$$= 18 \text{ kmh}^{-1} = \frac{5}{18} \times 18 = 5 \text{ ms}^{-1}$$

Therefore power developed by the horse = Force $\times$ velocity = $300 \times 5 = 1500 \text{ W}$

$$\text{Therefore equivalent horse power} = \frac{1500}{746} = 2.01 \text{ hp}$$

10. The boy has to overcome the force of gravity. Hence force of gravity on the boy

$$F = mg = 40 \times 9.8 = 392 \text{ N}$$

Total distance covered $s = 50 \times 10 = 500 \text{ cm} = 5 \text{ m}$

$$\text{i) Work done by the body in climbing} = \text{force} \times \text{distance} = W = 392 \times 5 = 1960 \text{ J}$$

$$\text{ii) Power developed} = \frac{W}{t} = \frac{1960}{5} = 392 \text{ W}$$

11. Because the person first moves a horizontal distance of 5 m. Hence for this part of motion the angle between the force and displacement is 90° .

$$\text{Hence, the work done is given by } W_1 = F \cos \theta = 10 \times \cos 90^\circ = (10) (5) (0) = 0$$

For the vertical motion, the angle between the force and the displacement is 0° . Therefore the work done is given by

$$W_1 = Fd \cos 0^\circ = (10)(10)(\cos 0^\circ) = (10)(10)(1) = 100 \text{ Nm} = 100 \text{ J}$$

$$\text{Hence, total work done} = W_1 + W_2 = 0 + 100 = 100 \text{ J}$$

12. All cutting instruments such as blades, axes etc. are sharpened so that the area of cross section decreases and hence pressure exerted by them increases. Thus they can easily penetrate a given surface.

13. Given : Power of bulb = 100 W = 100 Js⁻¹

$$\text{Time} = 2 \text{ hr} = 7200 \text{ s}$$

$$\text{Therefore, Energy consumed} = P \times t = 100 \times 7200 = 720,000 \text{ J}$$

14. Given, Mass, $m = 200 \text{ kg}$, $g = 10 \text{ m/s}^2$ and Distance, $h = 6 \text{ m}$

Work in lifting the mass is done against gravity.

$$\text{Therefore, the work done is } W = \text{Force} \times \text{Displacement} = mg \times h = mgh$$

$$W = mgh = 200 \times 10 \times 6 = 12000 \text{ J.}$$

15. In order to bring the object to rest the work done must be equal to the kinetic energy of the moving object = $\frac{1}{2}mv^2$

16. Here, Mass of Car = 150 Kg.

$$\text{Power developed by a car for one Kg} = 500 \text{ W}$$

$$\text{Total Power developed by a car for 150 Kg} = 150 \times 500 = 7.5 \times 10^4 \text{ W}$$

$$\text{We know that, Force} = \frac{\text{Power}}{\text{Velocity}} = \frac{7.5 \times 10^4}{20} = 3.75 \times 10^3 \text{ N}$$

$$\text{Therefore, Force} = 3750 \text{ N.}$$

17. Since Energy = power $\times$ time

$$= 4 \times 500 \times 10 = 20000 \text{ Wh} = 20 \text{ kWh}$$

18. Initial velocity = v , then final velocity, $v = 3v$

$$\text{Initial kinetic energy} = \frac{1}{2}mv^2$$

$$\text{Final kinetic energy (KE)} = \frac{1}{2}mv'^2 = \frac{1}{2}m(3v)^2 = 9\left(\frac{1}{2}mv^2\right)$$

$$(\text{KE})_{\text{initial}} : (\text{KE})_{\text{final}} = 1:9.$$

19. Given : $m = 1 \text{ kg}$, $h = 1 \text{ m}$, $g = 10 \text{ ms}^{-2}$ PE = ?

$$\text{Using } U = mgh = 1 \times 10 \times 1 = 10 \text{ J}$$

Since potential energy is converted completely into kinetic energy, therefore kinetic energy transferred to the nail is 10 J.

20. Here, Height, $h = 20 \text{ m}$, and mass of waterfall in a minute = 2000 tonn = $2000 \times 10^3 \text{ kg} = 2 \times 10^6 \text{ kg}$

$$\text{Hence, Power} = \frac{mgh}{t} = \frac{2 \times 10^6 \times 10 \times 20}{60} \text{ W} = \frac{4}{6} \times 10^7 \text{ W} = 6.6 \times 10^6 \text{ W.}$$

21. Given:

$$\text{i) Initial velocity (u) = 0,}$$

$$\text{Final velocity (v) = ?}$$

$$\text{Distance (s) = 50 cm = 0.5 m}$$

$$\text{Acceleration (g) = } 10 \text{ ms}^{-2}$$

Using the equation of motion,

$$v^2 - u^2 = 2gh$$

$$\Rightarrow v^2 = u^2 + 2gh$$

we have

$$V^2 = 0 + 2 \times 10 \times 0.5 = 10$$

$$v = \sqrt{10} = 3.16 \text{ ms}^{-1}$$

$$\text{ii) Kinetic energy of the body} = \frac{1}{2}mv^2 = \frac{1}{2} \times 10 \times 10 = 50 \text{ J}$$

iii) Velocity of the body does not depend upon its mass, because the earth attracts all bodies with same acceleration due to gravity.

22. The work needed in lifting a mass m to a height h against the gravitational force. $W = mgh$

$$\text{Power } P = \frac{W}{t}$$

$$\text{Given : } m = 10^5 \text{ kg, } g = 10 \text{ ms, } h = 360 \text{ m, } t = 1 \text{ h} = 60 \times 60 \text{ s} = 3600 \text{ s}$$

$$p = \frac{10^5 \times 10 \times 360}{3600} = 10^5 \text{ w}$$

23. When a machine or person does different amounts of work or uses energy in different interval of time, the ratio between the total work or energy consumed to the total time is average power.

$$\text{Average Power} = \frac{\text{Time work done or energy consumed}}{\text{Total time}}$$

24. Given : Mass of the rocket (m) = 3×10^6 kg. Height (h) = 25 km = 25×10^3 m

Acceleration due to gravity (g) = 10 ms^{-2}

P.E. of the rocket = mgh

$$= 3 \times 10^6 \times 10 \times 25 \times 10^3 = 7.5 \times 10^{11} \text{ J}$$

Now K.E. of the rocket

$$= \frac{1}{2}mv^2 = \frac{1}{2} \times 3 \times 10^6 \times 10^6 = 1.5 \times 10^{12} \text{ J}$$

25. The block of plastic displaces more weight of water than its own weight and therefore, experiences a buoyant force greater than its own weight. As a result the net force acts in the upward direction. Thus on being released it comes upto the surface of water.

26. Given mass, m = 500 kg

The given unit of speed is km/h. It is to be converted into m/s.

$$= 1 \text{ km/h}$$

$$= \frac{1 \times 1000 \text{ metre}}{3600 \text{ second}}$$

$$= \frac{5}{18} \text{ m/s}$$

Initial speed, u = 36 km/h

$$= 36 \times \frac{5}{18} \text{ m/s}$$

$$= 10 \text{ m/s}$$

Final speed = v = 72 km/h

$$= 72 \times \frac{5}{18} \text{ m/s}$$

$$= 20 \text{ m/s}$$

Gain in KE = Final KE - Initial KE

$$= \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

$$= \frac{1}{2}m \times (v^2 - u^2)$$

$$= \frac{1}{2} \times 500 \times [(20)^2 - (10)^2]$$

$$= \frac{1}{2} \times 500 \times [400 - 100]$$

$$= \frac{1}{2} \times 500 \times 300$$

$$= 75000 \text{ joule}$$

$$= 7.5 \times 10^4 \text{ J}$$

27. When a person holds a bundle of hay over his head for 30 minutes and gets tired he applies force in upward direction and displacement of bundle of hay is in forward direction which is perpendicular to the direction of force applied therefore displacement is zero

$$W = F \times s \cos \theta$$

$$= F \times s \cos 90^\circ$$

$$= F \times 0 = 0$$

No work done.

28. In order to balance the load on his head, the coolie applies a force on it in the upward direction, equal to its weight. His displacement is along the horizontal direction. Thus the angle between force F and displacement is 90° .

$$\text{Therefore, work done } W = Fs \cos \theta = Fs \cos 90^\circ = 0$$

29. No work is done, Since work done on the object by gravitational force depends upon change in the vertical height of the object.

Vertical height of the object is not changing as the joining A and B is horizontal at the same height hence the work done is zero.

30. No, Zero mechanical energy means that there is no potential energy and no kinetic energy. Therefore, if kinetic energy is zero, velocity becomes zero and hence, there will be no momentum.

31. Dead bodies always float on the surface of water, but the head stays within water. The reason is that when the dead body decays its volume increases. Thus, it becomes lighter than water and, hence, floats. However, head being heavy cannot displace water more than its own weight hence it remains under water.

32. i. Mass = 1g = 0.001 kg, speed = 0.5 m/s

$$P = \frac{W}{t} = \frac{mgh}{t} = \frac{0.001 \times 10 \times 0.5}{1} = 0.005 \text{ W}$$

ii. Mass = 250 g = 0.25 kg, velocity = 0.5 m/s

$$P = \frac{W}{t} = \frac{mgh}{t} = \frac{0.25 \times 10 \times 0.5}{1} = 1.25 \text{ W}$$

Power of squirrel is much higher than power of butterfly.

33. Electric heater's power (p) = 1500 W = 1.5 kW

$$\text{Energy} = \text{power} \times \text{time} = 1.5 \text{ kW} \times 10 \text{ h} = 15 \text{ kWh}$$

34. Energy consumed by a certain household = 250 kWh

since $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$

therefore $250 \text{ kWh} = 250 \times 3.6 \times 10^6 = 9 \times 10^8 \text{ J}$

35. Force of the gravity = mg (vertically downward everywhere)

Displacement of object AB (= s say) in horizontal direction.

Angle between force and displacement is 90°

$\therefore$ Work done, $W = Fs \cos = 90^\circ = 0$ (zero)

That is work done by gravity on projectile is zero.

36. $\text{K.E.} = \frac{1}{2} mv^2$

where K.E. - kinetic energy of an object

'm' - mass of an object

'v' - velocity

37. We have given that,

Mass of the car, $m = 1500 \text{ kg}$, the initial velocity of the car, $u = 30 \text{ km h}^{-1}$

$$u = \frac{30 \times 1000 \text{ m}}{60 \times 60 \text{ s}}$$

$$= \frac{25}{3} \text{ ms}^{-1}$$

Similarly, the final velocity of the car,

$$v = 60 \text{ km h}^{-1} = \frac{50}{3} \text{ ms}^{-1}$$

Therefore, the initial kinetic energy of the car,

$$E_{ki} = \frac{1}{2} mu^2$$

$$= \frac{1}{2} \times 1500 \text{ kg} \times \left(\frac{25}{3}\right)^2 \text{ ms}^{-1}$$

$$= \frac{156250}{3} \text{ J}$$

The final kinetic energy of the car

$$E_{kf} = \frac{1}{2} \times 1500 \text{ kg} \times \left(\frac{50}{3}\right)^2 \text{ ms}^{-1}$$

$$= \frac{625000}{3} \text{ J.}$$

Thus, the work done = Change in kinetic energy = $E_{kf} - E_{ki} = 156250 \text{ J}$.

38. Linear momentum of first object, $p_1 = m_1 v_1$ and of second object, $p_2 = m_2 v_2$

But, $p_1 = p_2$

or, $m_1 v_1 = m_2 v_2$

If $m_1 > m_2$ then $v_1 < v_2$

$$(K.E.)_1 = \frac{1}{2} (m_1 v_1) v_1 = \frac{1}{2} p_1 v_1$$

$$\text{and } (K.E.)_2 = \frac{1}{2} (m_2 v_2) v_2 = \frac{1}{2} p_2 v_2$$

$$\text{So, } \frac{(K.E.)_1}{(K.E.)_2} = \frac{\frac{1}{2} (p_1 v_1) v_1}{\frac{1}{2} (p_2 v_2) v_2} = \frac{v_1}{v_2}$$

But, $v_1 < v_2$

Therefore, $(K.E.)_1 < (K.E.)_2$

39. $K.E. = \frac{1}{2} (m \times v^2)$

Given initial kinetic energy = 25 J

given velocity

$$v = 5 \text{ m/s}$$

$$25 = \frac{1}{2} \times m \times 25$$

$$m = 2$$

When velocity is doubled, new velocity,

$$v = 10 \text{ m/s}$$

$$\text{K.E.} = \frac{1}{2} \times 2 \times (10^2)$$

$$K.E. = 100 \text{ J}$$

When velocity is made three times,

$$v = 15 \text{ m/s}$$

$$\text{K.E.} = \frac{1}{2} \times 2 \times (15^2)$$

$$= 225$$

40. Let the mass of each body be m . Then, K.E. of the first body, $K_1 = \frac{1}{2} mv^2$ (i)

$$\text{K.E. of the second body } K_2 = \frac{1}{2} mv_2^2 = \frac{1}{2} m(3v)^2 = \frac{9}{2} mv^2 \dots (ii)$$

$$\text{Dividing (ii) by (i) we have } \frac{K_2}{K_1} = \frac{\frac{9}{2} mv^2}{\frac{1}{2} mv^2} = \frac{9}{1}$$

Thus, the kinetic energy of the second body is nine times the kinetic energy of the first body.

41. Kinetic energy is the energy possessed by a body due to its motion. The two uses of kinetic energy are as follows:

- i. The kinetic energy of air is used to run windmills.
- ii. The kinetic energy of the running water is used to generate electricity.

42. Mass of object (m) = 40 kg

Height (h) = 5 m

Acc. due to gravity (g) = 10 ms^{-2}

$\therefore$ P.E. at a height of 5 m (PE) = mgh

$$= 40 \times 10 \times 5 = 2000 \text{ J}$$

P.E. at a half way height, i.e. : 2.5 m (PE)

$$= mgh$$

$$= 40 \times 10 \times 2.5$$

$$= 1000 \text{ J}$$

Decrease in P.E. = Increase in K.E.

$$= 2000 - 1000$$

$$= 1000 \text{ J}$$

43. A battery has stored chemical energy. The chemical energy is converted into electrical energy during lighting of bulb. So, a battery converts chemical energy into electrical energy. The filament in the bulb becomes white hot and gives out light. Electrical energy is converted into heat and light energy.

Thus, the energy changes involved in this process can be written as follows:

Chemical energy $\rightarrow$ Electrical energy $\rightarrow$ Heat energy + Light energy.

44. i. Energy consumed by five bulbs = $\frac{5 \times 100 \times 4}{1000} = 2$ units

Energy consumed by 1 heater = $\frac{1500 \times 2}{1000} = 3$ units

Energy consumed by electric iron = $\frac{1000 \times 5}{1000} = 5$ units

Total energy consumed by them = $(2 + 3 + 5) = 10$ units

ii. 1 unit or kWh = $3.6 \times 10^6 \text{ J}$

$$\therefore 10 \text{ units} = 3.6 \times 10^6 \times 10 \text{ J} = 3.6 \times 10^7 \text{ J}$$

45. Power of Avinash $P_A = F_A \cdot v_A = 10 \times 8 = 80 \text{ W}$

Power of Kapil $P_K = F_K \cdot v_K = 25 \times 3 = 75 \text{ W}$

So, Avinash is more powerful than Kapil.

46. Due to the presence of minerals, the density of sea water is more than the density of river water therefore upthrust is large. Hence, with a smaller portion submerged in sea water, the weight of sea water displaced is equal to the total weight of the body, while to displace the same weight of river water a large portion of the body will have to be submerged in water. Therefore it is easier to swim in sea water than in river water.

47. When the tin can is pushed into the water keeping its mouth upwards, it displaces more volume of water, and therefore it experiences more upthrust. But when it is pushed into the water with its mouth towards water, it displaces less volume of water (as water enters it). As a result the upthrust is also less. Thus it is comparatively easier to push a tin can into water with its mouth towards water than away from it.

48. i) Weight of hot air = Volume $\times$ Density $\times g$

$$= 50 \times 0.4 \times g$$

$$= 20 \text{ kgf}$$

ii) Weight of hot air, balloon and equipment

$$= 20 + 12 + P = (32 + P) \text{ kgf}$$

iii) Upthrust = Weight of air displaced

$$= h d g$$

$$= 50 \times 1.3 \times g$$

$$= 65 \text{ kgf}$$

By law of floatation we have,

$$32 + P = 65$$

$$P = 65 - 32 = 33 \text{ kgf}$$

49. Given, mass of A (m_A) = 60 kg

$$\text{mass of B } (m_B) = 40 \text{ kg}$$

$$\text{mass of luggage } (m_L) = 20 \text{ kg}$$

$$\text{Height of staircase } (h) = 0.5 \times 10 = 5 \text{ m}$$

$$\text{So, work done by boy A to climb staircase} = mgh = (60 + 20) \times 9.8 \times 5 = 3920 \text{ J}$$

$$\text{So, power of A} = \frac{\text{work}}{\text{time}} = \frac{3920}{20} = 196 \text{ W}$$

$$\text{Similarly, power of B} = \frac{\text{work}}{\text{time}} = \frac{mgh}{t}$$

$$\frac{[(40+20) \times 9.8 \times 5]}{10} = \frac{2940}{10} = 294 \text{ W}$$

i. B possesses greater power than A.

$$\text{ii. So, the ratio is given by } \frac{\text{power of A}}{\text{power of B}} = \frac{196}{294} = 2:3$$

$$\text{So, power, } P = \frac{W}{t} = \frac{3000}{3} = 1000 \text{ W}$$

50. Volume of water raised = 30,000 litres $\frac{30,000}{1000} \text{ m}^3 = 30 \text{ m}^3$.

$$\text{Mass of water raised, } M = \text{Volume} \times \text{Density}$$

$$= (30 \text{ m}^3) \times (10^3 \text{ kg/m}^3)$$

$$M = 3000 \text{ kg}$$

$$\text{Height, } h = 45 \text{ metre}$$

Work done by machine,

$$W = \text{Weight of water raised} \times \text{Height}$$

$$= (Mg) \times h = (30 \times 10^3 \times 9.8) \times 45$$

$$W = 1.323 \times 10^7 \text{ joule.}$$

$$\text{Time taken, } t = 10 \text{ minutes} = 10 \times 60 = 600 \text{ seconds.}$$

$$\text{Power, } P = \frac{W}{t}$$

$$\frac{1.323 \times 10^7 \text{ J}}{600 \text{ s}}$$

$$= 22 \times 10^3 \text{ W} = 22 \text{ kW.}$$

51. If a body of mass changes its velocity from "u" to "v" with an acceleration, "a" due to the application of a constant force, "F" in the direction of motion, then the displacement of the body, "S" is given by

$$v^2 = (u^2) + 2as$$

$$s = \frac{v^2 - u^2}{2a}$$

but the work done in changing the velocity of the body from u to v is

$$W = F.s$$

$$W = \frac{ma(v^2 - u^2)}{2a}$$

$$\Rightarrow W = \frac{m(v^2 - u^2)}{2}$$

$$W = \frac{mv^2}{2} - \frac{mu^2}{2}$$

But it is change in kinetic energy

Hence verified.

52. Energy consumed in a month = 250 units

$$= 250 \text{ kWh}$$

$$= 250 \text{ kW} \times 1 \text{ hr}$$

$$= 250 \times 1000 \text{ W} \times 3600 \text{ s}$$

$$= 900,000,000 \text{ J}$$

53. The ice berg has smaller density (0.917 g cm^{-3}) than sea water at the same temperature. The density of sea water is about 1.018 g cm^{-3} hence an iceberg floats with nearly one twelfth of its volume outside water.

54. Sheet crumpled into a ball has small surface area as compared to the similar unfolded sheet. Therefore unfolded sheet will experience more friction due to air as compared to the sheet crumpled into a ball in spite of same force of gravity acts upon them. It is larger friction of air which slows down the unfolded sheet and therefore it falls slower as compared to sheet crumpled into a ball.
55. i. Let us find Power expended by girl A:
 Weight of the girl, $mg = 400 \text{ N}$
 Displacement (height), $h = 8 \text{ m}$
 Time taken, $t = 20 \text{ s}$
 Power, $P = \frac{\text{Work done}}{\text{time taken}}$

$$= \frac{mgh}{t}$$

$$= \frac{400\text{N} \times 8\text{m}}{20\text{s}}$$

$$160 \text{ W}$$
- ii. Now, the power expended by girl B:
 Weight of the girl, $mg = 400 \text{ N}$ Displacement (height), $h = 8 \text{ m}$ Time taken, $t = 50 \text{ s}$
 Power, $P = \frac{mgh}{t}$

$$= \frac{400\text{N} \times 8\text{m}}{50\text{s}}$$

$$= 64 \text{ W.}$$

 The Power expended by girl A is 160 W.
 The Power expended by girl B is 64 W.
56. It is found that an iceberg floats in sea water with 90% of its volume below water and 10% of its volume above sea water. Icebergs are extremely dangerous for shipping as under water ice can hit the ship and sink it.
57. The rate of doing work is called power. Kilowatt is the unit of power and kilowatt-hour is the unit of energy. Calculation for power: Height = 20 m, mass = 2000 tonne = $2 \times 10^6 \text{ kg}$ and time = 1 min = 60s

$$\text{Power} = \frac{W}{t} = \frac{mgh}{t} = \frac{2 \times 10^6 \text{ kg} \times 10 \text{ m/s}^2 \times 20 \text{ m}}{60 \text{ s}}$$

$$= \frac{2}{3} \times 10^7 \text{ W}$$
58. Given : Mass, $m = 10 \text{ kg}$, Height, $h = 5 \text{ m}$, $g = 9.8 \text{ ms}^{-2}$
 Therefore Potential energy (P.E) at the top = $mgh = 10 \times 9.8 \times 5 \text{ m} = 490 \text{ J}$
 On reaching the ground the whole of the potential energy of the falling rock is converted into kinetic energy. Therefore Kinetic energy of the rock on reaching the ground = Potential energy of the rock at the top of the ladder
 Hence K.E. of the rock on the ground = 490 J
 Now Initial velocity, $u = 0 \text{ ms}^{-1}$, Final velocity, $v = ?$, Displacement, $s = 5 \text{ m}$ and $g = 9.8 \text{ ms}^{-2}$
 Using the relationship, $v^2 - u^2 = 2gS$, we have $v^2 - 0 = 2 \times 9.8 \times 5 = 98$
 Therefore $v = \sqrt{98} = 9.9 \text{ ms}^{-1}$
59. Given force, $F = 750 \text{ N}$, displacement = height,
 $h = 16 \text{ m}$ and time, $t = 5$
 Work done by machine,
 $W = \text{Force} \times \text{Displacement} = F \times h$
 $= 750 \text{ N} \times 16 \text{ joule}$
 Power of machine, $P = \frac{W}{t}$

$$\frac{750 \text{ N} \times 15 \text{ J}}{5 \text{ s}}$$

$$= 2400 \text{ watt.}$$
60. By definition: Work = Force $\times$ displacement in the direction of force = $Fs \cos \theta$
 Given : $F = 10 \text{ N}$, $s = 2 \text{ m}$, $\theta = 60^\circ$
 Therefore $W = 10 \times 2 \times \cos 60^\circ = 10 \times 2 \times \frac{1}{2} = 10 \text{ J}$
61. When a satellite moves around the Earth in a circular path, then the force of gravity acts on it directed towards the centre. The motion of the satellite is in the horizontal plane. Therefore, the force of gravity of Earth on the satellite and the direction of motion of satellite are perpendicular to each other. Therefore, net work done = $Fs \cos 90^\circ = 0$. That is, the work done by the force of gravity on a satellite moving around the Earth is zero.
62. Work done = force $\times$ displacement
 For calculating work done there must be displacement but here in this example vertical displacement is zero (as initial and final points lie on the same horizontal line) so the work done by force of gravity is zero.

63. $P = \sqrt{2mk}$. Since K is the same for both the bodies,
 $P \propto \sqrt{m}$ i.e. the heavier body has more momentum than the lighter body.
64. Kinetic energy is given by the relation $K = \frac{1}{2}mv^2$. Multiplying and dividing the right hand side of the equation by m we have $K = \frac{1}{2} \frac{mv^2}{m}$, but $mv = p$ is momentum of a body, therefore $p = \sqrt{2mk}$
65. i. The SI unit of power is watt.
 1 watt is the power of a body which does work at the rate of 1 joule per second.
 i.e. $1 \text{ watt} = \frac{1 \text{ joule}}{1 \text{ second}}$
- ii. Given, mass of one brick = 2.5 kg
 Mass of 10 bricks = $2.5 \times 10 = 25 \text{ kg}$
 Height, $h = 20 \text{ m}$, time, $t = 50 \text{ s}$, power, $P = ?$
 $\therefore \text{Power, } P = \frac{mgh}{t} = \frac{25 \times 10 \times 20}{50} = 100 \text{ Js}^{-1}$
66. We know that $KE \propto m$ and $KE \propto v^2$
 Therefore, by doubling the mass, the kinetic energy doubles, while by doubling the velocity, the kinetic energy increases four times. Therefore, doubling the velocity will have a greater effect on the kinetic energy of an object.
67. Given : Vertical height climbed = Height of step $\times$ No. of steps = $20 \text{ cm} \times 30 = 600 \text{ cm} = 6 \text{ m}$
 Acceleration due to gravity (g) = 10 ms^{-2}
 Mass of the boy (m) = 50 kg
 Since work done is equal to gain in potential energy, therefore, we have = $50 \times 10 \times 6 = 3000 \text{ J}$
 Hence power developed by the boy
 $P = \frac{w}{t} = \frac{3000}{30} = 100 \text{ w}$
68. In such a case when the density of solid is exactly equal to that of the liquid, it will remain in equilibrium and keep floating in it.
 The solid will float in such a way that the entire part of it will remain within the liquid, its upper surface coinciding with the liquid surface.
69. Yes, it is possible, if an object is moving in a circular path because force is always acting perpendicular to the direction of displacement.
70. Work done = Force $\times$ Displacement
 i.e. $W = F \times S$
71. Mass of car = 1500 kg
 Vel. of car = $60 \text{ kmh}^{-1} = \frac{60 \times 1000}{60 \times 60} = 16.67 \text{ ms}^{-1}$
 W.D. = Change in K.E. of the car = $K_f - K_i$
 $K_i = \frac{1}{2}mv^2 = \frac{1}{2} \times 1500 \times (16.67)^2$
 $= \frac{1}{2} \times 1500 \times 277.89$
 $= 208416.68$
 Also $K_f = 0$
 $\therefore W = 0 - 208416.68$
 $= -208416.68 \text{ J}$
72. In the straight up situation, the angle of inclination is more than the wind up situation. When mountain roads go wind up, the slope will be less than when the roads go straight up. So the frictional force is very large when the road winds up. So the tire of the vehicle will not slip easily when the roads go wind up. Another thing is that, in the straight up situation due to the large angle of inclination you have to work more to go up against your gravity, but when the roads are wind up due to the small angle of inclination you have to work less to go up against your gravity. That is why mountain roads rarely go straight up the slope but wind up gradually.
73. The mass of the balloon filled with helium is less than the mass of the air displaced by it. Hence upthrust acting on the balloon is more than its weight. As a result the balloon experiences a net upthrust which makes it rise. As the balloon rises it experiences lesser and lesser upthrust due to the fact that with height the density of air decreases. At a certain point the weight of the balloon may be completely balanced by the upthrust acting on it. Thereafter the balloon stops rising.
74. Here, Force (F) = 20 N , Displacement (S) = 1 m .
 Angle between force and displacement $\theta = 60^\circ$.
 $\therefore \text{Work done, } W = FS \cos \theta = 20 \times 1 \times \cos 60^\circ = 20 \times 1 \times \frac{1}{2} = 10 \text{ J. } (\cos 60^\circ = 1/2)$
 Therefore, The work done in displacing a given body through $1 \text{ m} = 10 \text{ J}$.

75. Mass of system, $m = \text{mass of girl} + \text{mass of trolley} = 35 + 5 = 40 \text{ kg}$.

Initial velocity, $u = 4 \text{ ms}^{-1}$, Final velocity, $v = 0$, Displacement, $S = 16 \text{ m}$

Using $v^2 - u^2 = 2aS$,

$$a = \frac{v^2 - u^2}{2S} = -\frac{16}{2 \times 16} = -\frac{1}{2} \text{ ms}^{-2}$$

$$\text{Force} = ma = 40 \times \left(-\frac{1}{2}\right) = -20 \text{ N}$$

1. Work done on the trolley = Force $\times$ Displacement = $20 \text{ N} \times 16 \text{ m} = 320 \text{ J}$.

2. Since, A girl is sitting on a trolley, Therefore, Work done by the girl = 0 J .